CSCE 883 — Midterm Formula Sheet

Suggested handwritten-sheet content · HW1–3 / lectures 1–9 · log = ln except entropy

Bayes / Naive Bayes / KNN

$$P(c \mid x) = \frac{P(x \mid c)P(c)}{\sum_k P(x \mid k)P(k)}.$$

NB assumes features independent *given class*:
$s_c = P(c)\prod_j P(x_j \mid c)$; $\hat{c} = \arg\max_c s_c$.
$P(c \mid x) = s_c / \sum_k s_k$; use log scores.
$P(c) = n_c/N$; categorical smoothing:

$$P(x_j = v \mid c) = \frac{n_{j,v,c} + 1}{n_c + V_j}.$$

Risk: $R(a \mid x) = \sum_c \lambda_{a,c} P(c \mid x)$.
Choose + if $P(+ \mid x) > \dfrac{C_{FP}}{C_{FP} + C_{FN}}$ (zero correct-class cost).
MLE: maximize likelihood. MAP: maximize likelihood × prior.
KNN: $d^2(x,q) = \sum_j (x_j - q_j)^2$.
Vote among K nearest; HW1 tie: nearest neighbor among tied classes.
Scale: $(x_j - \mu_{tr,j})/s_{tr,j}$ or
$(x_j - \min_{tr,j})/(\max_{tr,j} - \min_{tr,j})$.

MLE: write, log, differentiate, check

$L = \prod_i p(x_i \mid \theta)$; $\ell = \sum_i \log p(x_i \mid \theta)$. Solve $\ell' = 0$; check maximum and boundaries. Drop only parameter-independent terms.

Bernoulli / grouped HW2 PMF:
$\ell = A\log\theta + B\log(1-\theta) + C$
$\ell' = A/\theta - B/(1-\theta)$
$\hat{\theta} = A/(A+B)$; $\ell'' \le 0$.
HW2: $A = n_0 + n_1$, $B = n_2 + n_3$.

Laplace (location 0, scale b):
$p(x \mid b) = e^{-|x|/b}/(2b)$
$\ell = -N\log(2b) - \sum_i |x_i|/b$
$\ell' = -N/b + \sum_i |x_i|/b^2$

$$\boxed{\hat{b} = N^{-1}\sum_i |x_i|}.$$

Poisson: $p(x \mid \lambda) = \lambda^x e^{-\lambda}/x!$
$\ell = (\sum_i x_i)\log\lambda - N\lambda + C$
$\ell' = \sum_i x_i/\lambda - N$; $\boxed{\hat{\lambda} = \bar{x}}$.

Gaussian ($v = \sigma^2$):
$\ell = -\frac{N}{2}\log(2\pi v) - \frac{\sum_i (x_i-\mu)^2}{2v}$
$\partial_\mu \ell = \sum_i (x_i - \mu)/v$
$\partial_v \ell = -N/(2v) + \sum_i (x_i-\mu)^2/(2v^2)$

$$\boxed{\hat{\mu} = \bar{x}, \quad \hat{v} = \frac{1}{N}\sum_i (x_i - \bar{x})^2}.$$

MLE uses N; unbiased variance uses $N-1$.
Zero spread: no positive-variance maximizer; all-zero Laplace likewise for $b > 0$.

Gaussian regression NLL:
$\sum_i [\frac{1}{2}\log(2\pi) + \log\sigma_i + \frac{(y_i-\mu_i)^2}{2\sigma_i^2}]$.
If $s_i = \log\sigma_i^2$, variable part is
$\frac{1}{2}\sum_i [s_i + (y_i - \mu_i)^2 e^{-s_i}]$.
Fixed variance ⇒ squared error; learned variance requires log term.

Metrics / trees / evaluation

Rows=true, columns=predicted, order $[0, 1]$:

positive	matrix roles	
1	TN	FP
	FN	TP
0	TP	FN
	FP	TN

Acc $= (TP + TN)/N$
$P = TP/(TP + FP)$; $R = TPR = TP/(TP + FN)$
$F1 = 2TP/(2TP + FP + FN)$
$FPR = FP/(FP + TN)$; $TNR = 1 - FPR$.
Macro: average per-class metrics; macro-F1 averages *class F1s*. ROC: TPR vs FPR; use positive-class *scores*.
AUC $= P(s_+ > s_-) + \frac{1}{2}P(s_+ = s_-)$.
$MSE = \frac{1}{N}\sum_i (y_i - \hat{y}_i)^2$; $RMSE = \sqrt{MSE}$
$MAE = \frac{1}{N}\sum_i |y_i - \hat{y}_i|$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_{test})^2}.$$

Negative R^2: worse than test-set mean.
$H(S) = -\sum_c p_c \log_2 p_c$; $0\log 0 = 0$
$IG(S, A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v)$
Gini $= 1 - \sum_c p_c^2$. Pick max gain; recompute in each child. Pure: $H = 0$; balanced binary: $H = 1$. Unsplittable leaf: majority.
RF: bootstrap rows + random features per split. Cascade: append forest probabilities; use out-of-fold training predictions.

No leakage: split first; fit transforms on training features only. Tune/early stop on validation; freeze, then test. Refit transforms inside each CV fold. Group related rows if deployment needs unseen groups.

CNN: hand calculation + sizes

NN convention: **do not flip kernel**. For stride S, padding P (dilation 1):

$$Y_{oij} = b_o + \sum_{c,u,v} K_{ocuv} X_{c,iS+u-P,jS+v-P}.$$

Add bias once per cell, sum *all* input channels. Zero padding outside image. ReLU after sum; pooling after activation if specified.
Per spatial dimension, dilation D:

$$N_{out} = \left\lfloor \frac{N_{in} + 2P - D(K-1) - 1}{S} \right\rfloor + 1.$$

Apply to height and width separately.
Conv params (groups 1, biases):
$C_{out}(C_{in}K_H K_W + 1)$.
Per filter: $C_{in}K_H K_W$ weights + 1 bias.
Pool params: 0. Flatten: CHW.
Dense params: $(n_{in} + 1)n_{out}$.
HW3 shapes: $3\times24\times24 \to 12\times22\times22$
$\to 12\times11\times11 \to 128\times9\times9$
$\to 128\times4\times4 \to 2048 \to 10$.
Params: $336 + 13{,}952 + 20{,}490$.

Neural nets / backpropagation

$z^l = W^l a^{l-1} + b^l$; $a^l = \phi(z^l)$.
$\sigma(u) = (1 + e^{-u})^{-1}$; $\sigma' = \sigma(1-\sigma) \le 1/4$
$\tanh' = 1 - \tanh^2 u$; $\mathrm{ReLU}' = \mathbf{1}_{u>0}$
(at 0 choose subgradient, commonly 0).
$\phi_k(u) = \log(1 + e^{2ku})/(2k)$, $k > 0$
$\phi_k'(u) = \sigma(2ku)$; $k \to \infty$: ReLU.
BCE: $L = -y\log p - (1-y)\log(1-p)$,
$p = \sigma(z) \Rightarrow \boxed{\partial L/\partial z = p - y}$.
Half squared error: $L = \frac{1}{2}(p-y)^2$
$\Rightarrow \partial L/\partial z = (p-y)p(1-p)$.
Softmax: $p_k = e^{z_k}/\sum_j e^{z_j}$;
cross-entropy: $\partial L/\partial z_k = p_k - y_k$.
Subtract $\max z$ before exponentiating.

HW3 four-weight template:
$h_1 = \phi(w_1 x)$, $h_2 = \phi(w_2 x)$,
$z = w_3 h_1 + w_4 h_2$, $p = \sigma(z)$, $\delta = p - y$.
$\partial_{w_3} L = \delta h_1$; $\partial_{w_4} L = \delta h_2$
$\partial_{w_1} L = \delta w_3 \phi'(w_1 x) x$
$\partial_{w_2} L = \delta w_4 \phi'(w_2 x) x$.
$\boxed{w_i \leftarrow w_i - \eta \partial_{w_i} L}$: all gradients from *old* weights.
$\delta^l = ((W^{l+1})^T \delta^{l+1}) \odot \phi'(z^l)$
$\partial_{W^l} L = \delta^l (a^{l-1})^T$; $\partial_{b^l} L = \delta^l$.
Average gradients for mean batch loss.
Scalar sigmoid chain, $|w_l| \le 1$:
$|\partial h_L/\partial x| \le (1/4)^L$.
Residual $h' = h + F(h)$: $J = I + J_F$.
Gradient check:

$$\partial_i L \approx \frac{L(w + \epsilon e_i) - L(w - \epsilon e_i)}{2\epsilon}.$$

Tiny ϵ: roundoff; large ϵ: truncation.

Diffusion / GAN / sequence models

$\alpha_t = 1 - \beta_t$; $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

$$\boxed{x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon}$$

$\epsilon \sim \mathcal{N}(0, I)$; predict $\epsilon_\theta(x_t, t)$.
$L = \mathbb{E}\|\epsilon - \epsilon_\theta(x_t, t)\|^2$.
Elementwise mean: gradient $2(\hat{\epsilon} - \epsilon)/M$.

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

Train: random t, fixed noise schedule. Generate: start Gaussian, denoise iteratively.
GAN: $\min_G \max_D \{\mathbb{E}_x \log D(x)$
$+\mathbb{E}_z \log[1 - D(G(z))]\}$.
Minimize $L_D = -\mathbb{E}\log D(x) - \mathbb{E}\log(1 - D(G(z)))$;
non-saturating $L_G = -\mathbb{E}\log D(G(z))$.
Detach fake for D step; pass gradients through D for G. Mode collapse: low diversity.
RNN: $h_t = \phi(W_x x_t + W_h h_{t-1} + b)$.
LSTM: $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$;
$h_t = o_t \odot \tanh c_t$. Gates f, i, o: sigmoid; candidate $\tilde{c}$: tanh. Weights shared over time.

Code: zero gradients → forward → loss → backward → optimizer step. Eval disables dropout; no-grad disables gradient tracking.